

How Plants Get the Energy to Grow

Name _____

Date _____

Two Pots, One Difference

Two pots are set up to be compared. They hold identical soil, they are watered by the same amount, and they stand in the same room at the same temperature. The only difference between them is **where each pot sits**.



The two seedlings are treated the same in every way except where they stand.

A plant that adds height and unfolds new leaves is doing something that costs energy. That gives this lesson its question:

Where does the energy a plant uses to grow actually come from?

What you already know

Plants are living things that need light and water to grow.

Every living thing needs energy to stay alive.

Two possible answers

Where the growing energy comes from is exactly what people disagree about.

[illegible]

Oversized table continuation stress

TERM	MEANING
Row 1	Row 1 preserves the full table cell wording across page continuation. Row 1 preserves the full table cell wording across page continuation.
Row 2	Row 2 preserves the full table cell wording across page continuation. Row 2 preserves the full table cell wording across page continuation.
Row 3	Row 3 preserves the full table cell wording across page continuation. Row 3 preserves the full table cell wording across page continuation.
Row 4	Row 4 preserves the full table cell wording across page continuation. Row 4 preserves the full table cell wording across page continuation.
Row 5	Row 5 preserves the full table cell wording across page continuation. Row 5 preserves the full table cell wording across page continuation.
Row 6	Row 6 preserves the full table cell wording across page continuation. Row 6 preserves the full table cell wording across page continuation.
Row 7	Row 7 preserves the full table cell wording across page continuation. Row 7 preserves the full table cell wording across page continuation.
Row 8	Row 8 preserves the full table cell wording across page continuation. Row 8 preserves the full table cell wording across page continuation.
Row 9	Row 9 preserves the full table cell wording across page continuation. Row 9 preserves the full table cell wording across page continuation.

TERM	MEANING
Row 10	Row 10 preserves the full table cell wording across page continuation. Row 10 preserves the full table cell wording across page continuation.
Row 11	Row 11 preserves the full table cell wording across page continuation. Row 11 preserves the full table cell wording across page continuation.
Row 12	Row 12 preserves the full table cell wording across page continuation. Row 12 preserves the full table cell wording across page continuation.
Row 13	Row 13 preserves the full table cell wording across page continuation. Row 13 preserves the full table cell wording across page continuation.
Row 14	Row 14 preserves the full table cell wording across page continuation. Row 14 preserves the full table cell wording across page continuation.
Row 15	Row 15 preserves the full table cell wording across page continuation. Row 15 preserves the full table cell wording across page continuation.
Row 16	Row 16 preserves the full table cell wording across page continuation. Row 16 preserves the full table cell wording across page continuation.
Row 17	Row 17 preserves the full table cell wording across page continuation. Row 17 preserves the full table cell wording across page continuation.
Row 18	Row 18 preserves the full table cell wording across page continuation. Row 18 preserves the full table cell wording across page continuation.
Row 19	Row 19 preserves the full table cell wording across page continuation. Row 19 preserves the full table cell wording across page continuation.
Row 20	Row 20 preserves the full table cell wording across page continuation. Row 20 preserves the full table cell wording across page continuation.
Row 21	Row 21 preserves the full table cell wording across page continuation. Row 21 preserves the full table cell wording across page continuation.
Row 22	Row 22 preserves the full table cell wording across page continuation. Row 22 preserves the full table cell wording across page continuation.
Row 23	Row 23 preserves the full table cell wording across page continuation. Row 23 preserves the full table cell wording across page continuation.
Row 24	Row 24 preserves the full table cell wording across page continuation. Row 24 preserves the full table cell wording across page continuation.
Row 25	Row 25 preserves the full table cell wording across page continuation. Row 25 preserves the full table cell wording across page continuation.
Row 26	Row 26 preserves the full table cell wording across page continuation. Row 26 preserves the full table cell wording across page continuation.

TERM	MEANING
Row 27	Row 27 preserves the full table cell wording across page continuation. Row 27 preserves the full table cell wording across page continuation.
Row 28	Row 28 preserves the full table cell wording across page continuation. Row 28 preserves the full table cell wording across page continuation.
Row 29	Row 29 preserves the full table cell wording across page continuation. Row 29 preserves the full table cell wording across page continuation.
Row 30	Row 30 preserves the full table cell wording across page continuation. Row 30 preserves the full table cell wording across page continuation.
Row 31	Row 31 preserves the full table cell wording across page continuation. Row 31 preserves the full table cell wording across page continuation.
Row 32	Row 32 preserves the full table cell wording across page continuation. Row 32 preserves the full table cell wording across page continuation.
Row 33	Row 33 preserves the full table cell wording across page continuation. Row 33 preserves the full table cell wording across page continuation.
Row 34	Row 34 preserves the full table cell wording across page continuation. Row 34 preserves the full table cell wording across page continuation.
Row 35	Row 35 preserves the full table cell wording across page continuation. Row 35 preserves the full table cell wording across page continuation.
Row 36	Row 36 preserves the full table cell wording across page continuation. Row 36 preserves the full table cell wording across page continuation.
Row 37	Row 37 preserves the full table cell wording across page continuation. Row 37 preserves the full table cell wording across page continuation.
Row 38	Row 38 preserves the full table cell wording across page continuation. Row 38 preserves the full table cell wording across page continuation.
Row 39	Row 39 preserves the full table cell wording across page continuation. Row 39 preserves the full table cell wording across page continuation.
Row 40	Row 40 preserves the full table cell wording across page continuation. Row 40 preserves the full table cell wording across page continuation.
Row 41	Row 41 preserves the full table cell wording across page continuation. Row 41 preserves the full table cell wording across page continuation.
Row 42	Row 42 preserves the full table cell wording across page continuation. Row 42 preserves the full table cell wording across page continuation.
Row 43	Row 43 preserves the full table cell wording across page continuation. Row 43 preserves the full table cell wording across page continuation.

TERM	MEANING
Row 44	Row 44 preserves the full table cell wording across page continuation. Row 44 preserves the full table cell wording across page continuation.
Row 45	Row 45 preserves the full table cell wording across page continuation. Row 45 preserves the full table cell wording across page continuation.
Row 46	Row 46 preserves the full table cell wording across page continuation. Row 46 preserves the full table cell wording across page continuation.
Row 47	Row 47 preserves the full table cell wording across page continuation. Row 47 preserves the full table cell wording across page continuation.
Row 48	Row 48 preserves the full table cell wording across page continuation. Row 48 preserves the full table cell wording across page continuation.
Row 49	Row 49 preserves the full table cell wording across page continuation. Row 49 preserves the full table cell wording across page continuation.
Row 50	Row 50 preserves the full table cell wording across page continuation. Row 50 preserves the full table cell wording across page continuation.
Row 51	Row 51 preserves the full table cell wording across page continuation. Row 51 preserves the full table cell wording across page continuation.
Row 52	Row 52 preserves the full table cell wording across page continuation. Row 52 preserves the full table cell wording across page continuation.
Row 53	Row 53 preserves the full table cell wording across page continuation. Row 53 preserves the full table cell wording across page continuation.
Row 54	Row 54 preserves the full table cell wording across page continuation. Row 54 preserves the full table cell wording across page continuation.
Row 55	Row 55 preserves the full table cell wording across page continuation. Row 55 preserves the full table cell wording across page continuation.
Row 56	Row 56 preserves the full table cell wording across page continuation. Row 56 preserves the full table cell wording across page continuation.
Row 57	Row 57 preserves the full table cell wording across page continuation. Row 57 preserves the full table cell wording across page continuation.
Row 58	Row 58 preserves the full table cell wording across page continuation. Row 58 preserves the full table cell wording across page continuation.
Row 59	Row 59 preserves the full table cell wording across page continuation. Row 59 preserves the full table cell wording across page continuation.
Row 60	Row 60 preserves the full table cell wording across page continuation. Row 60 preserves the full table cell wording across page continuation.

TERM	MEANING
Row 61	Row 61 preserves the full table cell wording across page continuation. Row 61 preserves the full table cell wording across page continuation.
Row 62	Row 62 preserves the full table cell wording across page continuation. Row 62 preserves the full table cell wording across page continuation.
Row 63	Row 63 preserves the full table cell wording across page continuation. Row 63 preserves the full table cell wording across page continuation.
Row 64	Row 64 preserves the full table cell wording across page continuation. Row 64 preserves the full table cell wording across page continuation.
Row 65	Row 65 preserves the full table cell wording across page continuation. Row 65 preserves the full table cell wording across page continuation.
Row 66	Row 66 preserves the full table cell wording across page continuation. Row 66 preserves the full table cell wording across page continuation.
Row 67	Row 67 preserves the full table cell wording across page continuation. Row 67 preserves the full table cell wording across page continuation.
Row 68	Row 68 preserves the full table cell wording across page continuation. Row 68 preserves the full table cell wording across page continuation.
Row 69	Row 69 preserves the full table cell wording across page continuation. Row 69 preserves the full table cell wording across page continuation.
Row 70	Row 70 preserves the full table cell wording across page continuation. Row 70 preserves the full table cell wording across page continuation.
Row 71	Row 71 preserves the full table cell wording across page continuation. Row 71 preserves the full table cell wording across page continuation.
Row 72	Row 72 preserves the full table cell wording across page continuation. Row 72 preserves the full table cell wording across page continuation.
Row 73	Row 73 preserves the full table cell wording across page continuation. Row 73 preserves the full table cell wording across page continuation.
Row 74	Row 74 preserves the full table cell wording across page continuation. Row 74 preserves the full table cell wording across page continuation.
Row 75	Row 75 preserves the full table cell wording across page continuation. Row 75 preserves the full table cell wording across page continuation.
Row 76	Row 76 preserves the full table cell wording across page continuation. Row 76 preserves the full table cell wording across page continuation.
Row 77	Row 77 preserves the full table cell wording across page continuation. Row 77 preserves the full table cell wording across page continuation.

TERM	MEANING
Row 78	Row 78 preserves the full table cell wording across page continuation. Row 78 preserves the full table cell wording across page continuation.
Row 79	Row 79 preserves the full table cell wording across page continuation. Row 79 preserves the full table cell wording across page continuation.
Row 80	Row 80 preserves the full table cell wording across page continuation. Row 80 preserves the full table cell wording across page continuation.
Row 81	Row 81 preserves the full table cell wording across page continuation. Row 81 preserves the full table cell wording across page continuation.
Row 82	Row 82 preserves the full table cell wording across page continuation. Row 82 preserves the full table cell wording across page continuation.
Row 83	Row 83 preserves the full table cell wording across page continuation. Row 83 preserves the full table cell wording across page continuation.
Row 84	Row 84 preserves the full table cell wording across page continuation. Row 84 preserves the full table cell wording across page continuation.
Row 85	Row 85 preserves the full table cell wording across page continuation. Row 85 preserves the full table cell wording across page continuation.
Row 86	Row 86 preserves the full table cell wording across page continuation. Row 86 preserves the full table cell wording across page continuation.
Row 87	Row 87 preserves the full table cell wording across page continuation. Row 87 preserves the full table cell wording across page continuation.
Row 88	Row 88 preserves the full table cell wording across page continuation. Row 88 preserves the full table cell wording across page continuation.
Row 89	Row 89 preserves the full table cell wording across page continuation. Row 89 preserves the full table cell wording across page continuation.
Row 90	Row 90 preserves the full table cell wording across page continuation. Row 90 preserves the full table cell wording across page continuation.
Row 91	Row 91 preserves the full table cell wording across page continuation. Row 91 preserves the full table cell wording across page continuation.
Row 92	Row 92 preserves the full table cell wording across page continuation. Row 92 preserves the full table cell wording across page continuation.
Row 93	Row 93 preserves the full table cell wording across page continuation. Row 93 preserves the full table cell wording across page continuation.
Row 94	Row 94 preserves the full table cell wording across page continuation. Row 94 preserves the full table cell wording across page continuation.

TERM	MEANING
Row 95	Row 95 preserves the full table cell wording across page continuation. Row 95 preserves the full table cell wording across page continuation.
Row 96	Row 96 preserves the full table cell wording across page continuation. Row 96 preserves the full table cell wording across page continuation.
Row 97	Row 97 preserves the full table cell wording across page continuation. Row 97 preserves the full table cell wording across page continuation.
Row 98	Row 98 preserves the full table cell wording across page continuation. Row 98 preserves the full table cell wording across page continuation.
Row 99	Row 99 preserves the full table cell wording across page continuation. Row 99 preserves the full table cell wording across page continuation.
Row 100	Row 100 preserves the full table cell wording across page continuation. Row 100 preserves the full table cell wording across page continuation.
Row 101	Row 101 preserves the full table cell wording across page continuation. Row 101 preserves the full table cell wording across page continuation.
Row 102	Row 102 preserves the full table cell wording across page continuation. Row 102 preserves the full table cell wording across page continuation.
Row 103	Row 103 preserves the full table cell wording across page continuation. Row 103 preserves the full table cell wording across page continuation.
Row 104	Row 104 preserves the full table cell wording across page continuation. Row 104 preserves the full table cell wording across page continuation.
Row 105	Row 105 preserves the full table cell wording across page continuation. Row 105 preserves the full table cell wording across page continuation.
Row 106	Row 106 preserves the full table cell wording across page continuation. Row 106 preserves the full table cell wording across page continuation.
Row 107	Row 107 preserves the full table cell wording across page continuation. Row 107 preserves the full table cell wording across page continuation.
Row 108	Row 108 preserves the full table cell wording across page continuation. Row 108 preserves the full table cell wording across page continuation.
Row 109	Row 109 preserves the full table cell wording across page continuation. Row 109 preserves the full table cell wording across page continuation.
Row 110	Row 110 preserves the full table cell wording across page continuation. Row 110 preserves the full table cell wording across page continuation.
Row 111	Row 111 preserves the full table cell wording across page continuation. Row 111 preserves the full table cell wording across page continuation.

TERM	MEANING
Row 112	Row 112 preserves the full table cell wording across page continuation. Row 112 preserves the full table cell wording across page continuation.
Row 113	Row 113 preserves the full table cell wording across page continuation. Row 113 preserves the full table cell wording across page continuation.
Row 114	Row 114 preserves the full table cell wording across page continuation. Row 114 preserves the full table cell wording across page continuation.
Row 115	Row 115 preserves the full table cell wording across page continuation. Row 115 preserves the full table cell wording across page continuation.
Row 116	Row 116 preserves the full table cell wording across page continuation. Row 116 preserves the full table cell wording across page continuation.
Row 117	Row 117 preserves the full table cell wording across page continuation. Row 117 preserves the full table cell wording across page continuation.
Row 118	Row 118 preserves the full table cell wording across page continuation. Row 118 preserves the full table cell wording across page continuation.
Row 119	Row 119 preserves the full table cell wording across page continuation. Row 119 preserves the full table cell wording across page continuation.
Row 120	Row 120 preserves the full table cell wording across page continuation. Row 120 preserves the full table cell wording across page continuation.

OPTION 1

Comparison 1

Comparison card 1 continues at safe paragraph boundaries with all words retained. Comparison card 1 continues at safe paragraph boundaries with all words retained. Comparison card 1 continues at safe paragraph boundaries with all words retained. Comparison card 1 continues at safe paragraph boundaries with all words retained. Comparison card 1 continues at safe paragraph boundaries with all words retained. Comparison card 1 continues at safe paragraph boundaries with all words retained. Comparison card 1 continues at safe paragraph boundaries with all words retained. Comparison card 1 continues at safe paragraph boundaries with all words retained. Comparison card 1 continues at safe paragraph boundaries with all words retained. Comparison card 1 continues at safe paragraph boundaries with all words retained.

OPTION 2

Comparison 2

Comparison card 2 continues at safe paragraph boundaries with all words retained. Comparison card 2 continues at safe paragraph boundaries with all words retained. Comparison card 2 continues at safe paragraph boundaries with all words retained. Comparison card 2 continues at safe paragraph boundaries with all words retained. Comparison card 2 continues at safe paragraph boundaries with all words retained. Comparison card 2 continues at safe paragraph boundaries with all words retained. Comparison card 2 continues at safe paragraph boundaries with all words retained. Comparison card 2 continues at safe paragraph boundaries with all words retained. Comparison card 2 continues at safe paragraph boundaries with all words retained. Comparison card 2 continues at safe paragraph boundaries with all words retained.

[illegible][illegible]

[illegible]

A diagram showing a linear chain of five boxes labeled A, B, C, D, and E. Box A is connected to B, B to C, C to D, and D to E. Box E is also connected to C.

$$\boxed{F} + \boxed{G} + \boxed{H} + \boxed{I}$$

- ---

Use the evidence to identify the result.

1. Read the table row by row.
2. Preserve each term.

∴ Answer: The result follows from every retained term.

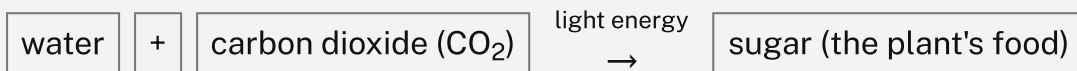
Where the Growing Energy Comes From

Key idea

The energy a plant grows on is captured from light, not taken in from the soil.

The green parts of a plant — mostly its leaves — absorb **light energy**. The leaf uses that energy to make the plant's own **food**: a sugar built from water drawn up out of the soil and carbon dioxide taken in from the air.

In one line



The energy stored inside that sugar is what the plant spends on building new leaves, stem and roots.

Follow the energy, step by step

Sunlight reaches the leaf.

The leaf's green parts capture that light energy.

The captured energy is used to build sugar from water and carbon dioxide. The energy is now stored inside food the plant has made for itself.

The plant breaks some of that sugar down again and uses the released energy to build new leaves, stem and roots. That is growth.

Notice two things

The energy enters the plant through the leaf, never through the soil. The only things arriving from the soil are water and dissolved minerals. Leaf area can be measured in m².

So what is the soil for?

Soil does a real job. It holds the plant firmly in place, and it holds water and dissolved minerals, which the roots take in and which then travel up to the leaves. Minerals are raw materials. They are not food, and they carry no usable energy.

WHAT THE PLANT TAKES IN	WHERE FROM	IS IT ENERGY?
Water	Soil, through the roots	No. A raw material for making sugar.
Dissolved minerals	Soil, through the roots	No. Raw materials for building the plant's body.
Light	Captured by the leaves	Yes. It ends up stored in sugar.

So a plant standing in rich soil with plenty of water, but kept in the dark, cannot make its own food and cannot grow, however well watered it is.

Two Beliefs That Fail the Evidence

Common belief 1

The belief *"The soil feeds the plant. A plant kept in the dark will keep growing as long as it is watered."*

The evidence Both pots held the same soil and got the same water. If the soil were supplying food and energy, the cupboard plant should have grown just as well. It stopped growing and turned pale.

So Soil supplies only water and dissolved minerals. The plant's food is made in its own leaves, using light energy.

Why did the cupboard plant keep going for a while? It was living on food it had already made and stored. Once that store ran down, it stopped growing and lost its green colour.

Common belief 2

The belief "Light only keeps a plant warm or makes it look healthy. The energy for growth comes from the water and the minerals."

The evidence Both pots stood in the same room at the same temperature. The cupboard was not cold, so the cupboard plant lacked no warmth. It still did not grow.

So Light is not merely warmth. The leaves capture light energy and use it to make the plant's own food, and growth runs on the energy stored in that food.

Which Observation Challenges the Soil Idea?

Plants get their food from the soil, so a plant in a dark room will keep growing as long as I water it.

A student says

That claim is wrong. The strong move is not to repeat the explanation, but to point to **evidence the claim cannot explain**.

Which observation would best challenge the student's idea?

- A.** A watered plant kept in darkness becomes pale and stops growing larger.
- B.** A watered plant in darkness grows taller because it stretches toward the light.
- C.** A watered plant in darkness stays cool, which proves it lacks growth energy.
- D.** A watered plant in darkness grows well when it is given plant food.